Cyber Security:

Cyber Security is a practice of protecting the systems, networks and programs from the digital attacks. These digital attacks are usually aimed to accessing, changing and destroying the sensitive data and they blackmail the money from user or interrupting normal business process using ransomware and other techniques

Cyber means digital or virtual world

Security means protecting the data

Cyber Security means protecting data in digital world

Hacker/Attacker/Hijacker:

- ➔ Hacker is a person who using different types of technical knowledge and other skill to gain illegal and unauthorized access on data over internet.
- ➔ In cyber security there are three of hackers are there they are
  - ❖ Black Hat Hacker
  - ❖ Gray Hat Hacker
  - ❖ White Hat Hacker

Black Hat Hacker:

- ➔ Black hat hacker is a person who breaks the systems and network by applying malicious codes and harmful instructions to breach the security and gain the access over data.
- ➔ Black hat hackers are very dangerous attackers in network.

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Gray Hat Hackers:

➔ Gray hat hackers are same as block hat hackers but they never destroy system or network resources just they are performing hacking techniques for making funny

White Hat Hackers:

➔ White hat hackers are very good persons and also call them as cyber security professionals and they are working under public and private organizations to find, analyze and pix problems from the attackers

➔ White hat hackers are called as **Ethical Hackers** in cyber security.

Types of Securities

➔ There are two types of securities in cyber world

1. Offensive Security
2. Defensive Security

Offensive Security:

➔ Offensive security is a type of cyber security in which ethical hacker can simulate real time attacks on systems, networks or applications to find and fix the vulnerabilities before malicious hackers exploit them.

Skills required becoming offensive security professionals

- Network Knowledge (TCP/IP, OSI Model, Firewall, etc…)
- Operating System like Linux/Windows
- Programming & Scripting (Python, Java, Bash, PowerShell etc)
- Familiarity with security tools (Kali Linux, Wireshark)
- Knowledge of attack techniques (fishing, Privilige Escalation etc).

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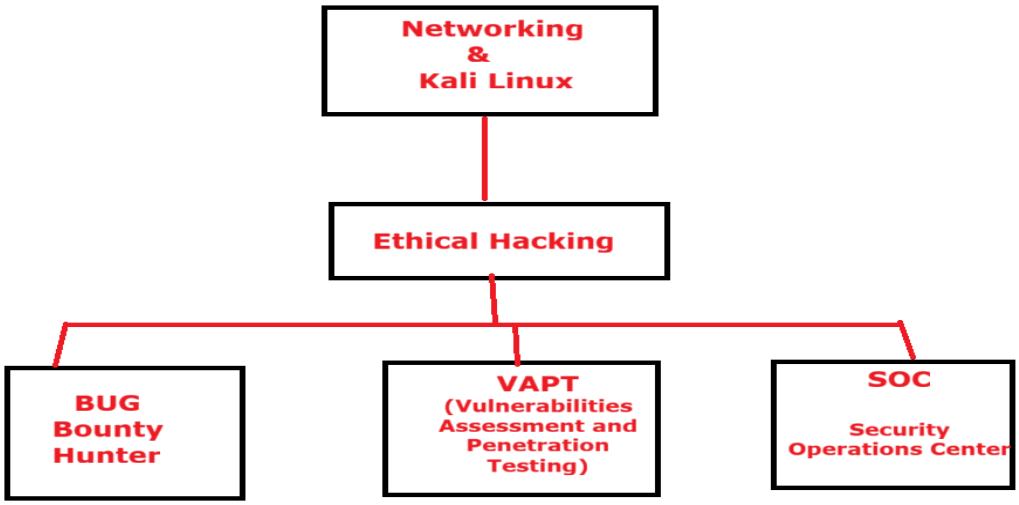
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Defensive Security:

➔ Defensive Security is a process of protecting systems, networks and data from cyber attacks by using tools, polices, strategies to detect and prevent the attacks

Road map for cyber Security



Ethical Hacking

➔ Ethical hacking is a process of legally and safely testing computers, software's, websites, networks or applications to find fix the security weaknesses before malicious hackers exploit them.

➔ White hat hackers are called as ethical hacker they can use tools and techniques as cybercriminals by with prior permissions for defensive purposes

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Bug Bounty Hunter:

- ➔ Bug bounty hunter is a ethical hacker who find the vulnerabilities in software's, websites, networks or applications and report them to the organization. In return, organization gives bounty (cash rewards or prizes) to appreciate them.

VAPT (Vulnerabilities Assessment and Penetration Testing):

- ➔ VAPT is also called as ethical hacker and he can work on behalf of organizations as cyber security professional and they can perform all the activities like offensive security

SOC (Security Operational Centers):

- ➔ SOC also called as ethical hacker and they can work like defensive cyber security professionals

Attack:

- ➔ An attack is any unauthorized attempt to access or steal or damage or disrupt the computer systems, networks or data
- ➔ If the attack is launched successfully, the affection of attack would be a loss of data or corruption of data or injection of viruses, warms, or malicious software's in to the network or defacing servers or many affections are happens.
- ➔ These attacks can be two types
    - ❖ Passive Attacks
    - ❖ Active Attacks

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Passive Attacks:

- ➔ Passive attacks are attempts to learn and make use of information that is being transmitted from one user to another user over internet
- ➔ The main intention of passive attacker is just performing action to attempt unauthorized reading or seeing messages and no modification of message involved.
- ➔ Thus **system resources** are does not affect in passive attacks
- ➔ These passive attacks mainly used for eavesdropping or monitoring transmission of message.
- ➔ There are two ways we can do passive attacks
  - ❖ Release of message content
  - ❖ Traffic Analysis

Active Attacks:

- ➔ In active attacks, attacker can modify the data that is being transmitted from one sender to receiver also attacker create false communication stream between sender and receiver.
- ➔ Active attacks can be four types
  - ❖ Masquerade
  - ❖ Reply Attack
  - ❖ Modification of message.
  - ❖ Denial of Service (DoS)

Cyber Attacks:

- ➔ A cyber attack is a malicious attempt by hacker or unauthorized user to damage or steel the computer or networks or a data

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Malware:

- ➔ Malware is any malicious software or any program that is intentionally designed to harmful to the computers, servers and networks to leak private information and gain the unauthorized access over the data and computers
- ➔ In cyber security we have different types of malwares are there,
  - ❖ Spyware
  - ❖ Virus
  - ❖ Worms
  - ❖ Trojan Horses
  - ❖ Ransomware
  - ❖ SQL Injection

Spyware:

- ➔ Spyware is a type of malware software that gathers the sensitive information from the user and secretly monitor the computers and users activities such as keystrokes, browsing habits and login credentials and send this data to the attackers.

Virus:

- ➔ A virus is a type of malicious software that comes through attached files or programs physically and it spread into the system whenever that file or program is run.
- ➔ Once virus is entered in to the system it corrupt or delete the files and spread into the other file systems and reducing the system performance.

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Worm:

- ➔ Worm is same as virus but it don't require any files or programs but it spreads automatically when we connected to network or wifi.
- ➔ Once we connect to the illegal network or wife the worm virus automatically creates its own replicas and spreads them over entire system.

Difference between Virus and worms

Virus	Worms
1. It is a malicious program that needs a host file to spread	1. It is self replicating malicious program that spreads without host file.
2. It spreads when the program is executed by the user	2. It spreads automatically over network by exploiting Vulnerablillties
3. Ex: ILOVEYOU, Melissa	3. Ex: Morris Worm, WannaCry
4. It needs a file + user action to spread	4. Worm spread automatically through networks

Trojan horse:

- ➔ Trojan horse is a type of malware that seems to be legitimate software or file or program to scam the users and it force to the user to install it.
- ➔ Once installed, it can give hackers control over systems or help spread further attacks within the network.

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Ransom ware:

- ➔ Ransomware is a type of malware by which attacker can block the access of the computer or a server until user paid ransom (crypto currency) payment.

Pretending:

- ➔ In cyber security, pretending means the attackers can act as someone else or something else as real to trick the people into giving them their access of the sensitive information like password, bank details, credit card details, debit card details or money transfer etc.

There are different types of attacks comes under pretending as given below

- ❖ Phishing
- ❖ Spoofing
- ❖ Social Engineering
- ❖ Pharming

Phishing Attack:

- ➔ Phishing is a type of cyber crime where cyber criminals send a fake emails or messages pretending to be from a trusted organization like bank, government office or a company.
- ➔ The aim of phishing attack is to steal the personal details, bank information and password by tricking people into clicking them on those fake links.

  Ex: A person receives a message **Congratulation!** You have won Rs. 10 Lakhs click here to link to claim. Existed, they click the link automatically it is asking bank account details. After entered the account details your account will be blocked very soon.

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Types of phishing attacks:

- Email Phishing -> Sending fake emails

  Ex: ceo@company.com (real email id)

  ceo@cornpany.com (fake email id)
- Vishing -> Sending fake links over phone calls or voice call
- Smishing -> Sending fake links over text messages (SMS)

Spoofing:

➔ Spoofing is one type of attack where attackers pretending to be a trusted network or IP address to steal the sensitive data from the computers

Social Engineering:

➔ Social Engineering is another type of attack where attackers pretending to be a trusted authority to manipulate the peoples to send their sensitive information.

Pharming:

➔ Pharming is another type of attack where attacker can create fake websites seems to be real website to hack the credentials.

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